

FYP: IoV Alert System Complete Flow

1📷 Camera Setup

- Camera khareedna (USB ya IP camera)
- Camera ko edge device network se connect karna
- Camera feed laptop / edge device me available hona chahiye

2📺 Edge Device (Laptop)

- Trained AI model run karna (VS Code / Python)
- Real-time video feed process karna → abnormality detect karna (fighting, murder etc.)
- Detection pe alert JSON generate karna

Example alert JSON:

```
{
  "camera_id": "CAM_01",
  "event": "Fighting",
  "confidence": 92,
  "location": {"lat": 31.45, "lng": 74.37},
  "status": "new",
  "assigned_to": null
}
```

3📶 Edge Device → Cloud / Firebase

- Alert cloud / Firebase server pe push karna
- Optional: small cropped video frame bhejna for verification
- Alert status = 'new'

4☁ Cloud → Auto-Assign Nearest IoV

- Cloud function trigger on new alert
- IoVs GPS location check karna
- Nearest IoV assign karna → update 'assigned_to' field
- Status update = 'assigned'

5? IoV Flutter App

- IoV app login hona → Firebase listener active
- Only alerts assigned_to == myID receive hong
- Push notification + sound + vibration
- Map location show karna
- Buttons: Accept / Reject

6? IoV Accepts Alert

- App alert status update kare: 'accepted'
- Web dashboard / Cloud dikhe: 'Alert accepted by IoV_123'
- Alert IoV app se remove ho jaye

7? Web Dashboard (Optional)

- Real-time multi-camera monitoring
- Alert history + severity
- IoV assignment / status visualization

8? Optional Advanced

- Multiple cameras → multiple edge devices
- Nearby IoV auto-assignment
- Alert timeout → auto reassign if IoV does not accept
- Model optimization for faster inference

9? Networking Points

- Edge → Camera: LAN / WiFi (RTSP)
- Edge → Cloud: Internet (HTTPS / Firebase SDK)
- IoV App → Cloud: Internet (Real-time listener / push notifications)
- Web Dashboard → Cloud: Internet (HTTP/HTTPS, WebSockets optional)

Simple Flow Diagram

Camera → Edge Device (Laptop with AI model)

→ Firebase / Cloud (Alert JSON)

→ Cloud Function → Assign Nearest IoV

→ IoV Flutter App → Push Notification

→ Web Dashboard → Monitoring + Alert Status

Summary: Camera → Laptop AI → Cloud → Nearest IoV → Alert → Accept → Web Dashboard update

Ye complete FYP flow professional, real-time aur scalable setup ke liye hai.